

AI Agents for High-Volume Customer Support

This document contains three case studies focused on designing and understanding AI Agents for High-Volume Customer Support.

Instructions

Please complete 3 case studies as per the guidelines below. Upload your completed case study solutions to SharePoint and share the link in the form: [Case Study Solution Submission – Fill out form](#)

Case Study 1: AI Customer Support Agent for E-Commerce Platforms

Executive Summary

E-commerce platforms receive extremely high volumes of customer queries related to orders, returns, refunds, and delivery tracking—especially during peak sales periods. An AI Customer Support Agent is introduced to handle these interactions efficiently.

Business Problem

Support teams struggle to manage repetitive customer queries during traffic spikes, leading to long wait times, increased operational costs, and reduced customer satisfaction.

Objectives

- Automate high-frequency customer support queries
- Reduce response times during peak loads
- Improve first-contact resolution
- Ensure smooth escalation to human agents

Solution Overview

The AI agent uses an LLM with Retrieval-Augmented Generation (RAG) to answer FAQs, retrieve order information, and initiate simple workflows such as returns.

Workflow

1. Customer submits a query via chat
2. Agent identifies intent and key entities
3. Relevant knowledge or order data is retrieved
4. Response is generated or action is triggered
5. Complex issues are escalated with context

Sample Q&A

Q: Where is my order? It was supposed to arrive yesterday.

A: Your order is currently delayed due to logistics issues and is expected to arrive tomorrow. Would you like to receive delivery updates?

Q: I want to return the item I received.

A: I can help you initiate a return. Please share your order ID to proceed.

Case Study 2: AI Customer Support Agent for Banking using GitHub Copilot

Executive Summary

Banks handle thousands of daily customer queries related to accounts, transactions, and cards. An AI Customer Support Agent is developed to manage this volume, with GitHub Copilot accelerating the development of secure backend services.

Business Problem

Manual handling of repetitive banking queries results in high call-center costs and slower customer service.

Objectives

- Automate Tier-0 and Tier-1 banking support queries
- Maintain compliance and security standards
- Reduce dependency on human agents
- Speed up development using AI coding assistance

Role of GitHub Copilot

GitHub Copilot assists developers by generating API code, suggesting secure patterns, and helping write validation logic and tests for customer support workflows.

Workflow

1. Customer initiates a secure chat session
2. Agent classifies intent and evaluates risk
3. Knowledge-based response or validation step is executed
4. Sensitive requests are escalated if required

Sample Q&A

Q: Why is my card transaction still pending?

A: Pending transactions are temporary holds by merchants and usually complete within a few business days.

Q: My debit card is blocked.

A: For security reasons, I need to verify your identity before helping you unblock the card or escalating this request.

Case Study 3: AI Customer Support Agent using Google ADK and LangChain

Executive Summary

High-volume enterprise customer support often requires handling complex, multi-step issues. This case study presents a modular AI Customer Support Agent built using Google ADK and LangChain.

Business Problem

Traditional single-agent chatbots struggle with scalability, context handling, and complex issue resolution.

Objectives

- Support large volumes of concurrent conversations
- Improve accuracy through retrieval-based responses
- Enable structured escalation flows
- Maintain conversation context across sessions

Role of Google ADK and LangChain

Google ADK manages agent teams, execution flow, and session state. LangChain enables retrieval, tool usage, and reasoning for customer support interactions.

Workflow

1. Customer submits a support request
2. Planner agent routes the request
3. LangChain retrieves relevant knowledge
4. Response is generated or escalated

Sample Q&A

Q: My application fails every time I try to log in.

A: I will check known issues and recent incidents related to login failures before suggesting next steps.

Q: This issue is still not resolved.

A: I am escalating this issue to a support specialist with the full conversation context and diagnostics.